

Junjie Xu

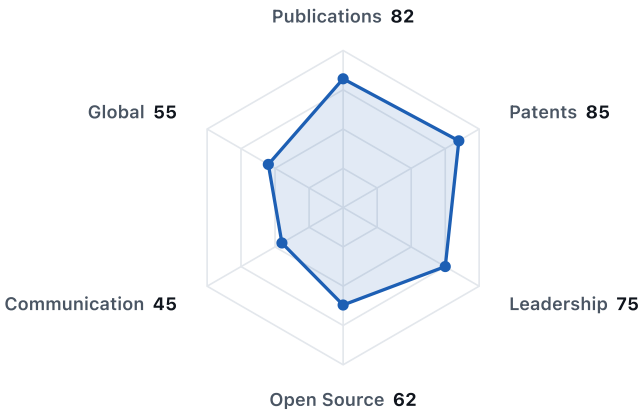
Direct Ph.D. Researcher · Affective AI · East China Normal University

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RESEARCH INTERESTS

Multimodal Affective Computing · Cross-modal Alignment · Music-to-Image Generation · Human-AI Collaboration

COMPETENCY RADAR



Strengths

Patents & Innovation · 85

14 published Chinese patent applications as co-inventor.

Research Output · 82

15 first-author papers across IPM, TCSVT, TALLIP, ACM MM workshop, and a CSUR survey.

Project Leadership · 75

Led two funded projects; both completed with excellent ratings.

Self-assessment, reviewed quarterly. The lowest dimensions point to where to improve next.

EDUCATION

2023 — 2028 (expected)	Direct Ph.D. student, Computer Science — East China Normal University, Shanghai
2026.11 — 2027.11	Visiting researcher, Nanyang Technological University — hosted by Prof. Eric Cambria (CSC-funded)

PUBLICATIONS

First-author publications; also contributing to multimodal sentiment analysis, visual question answering, and preference learning.

APEX: Learning Adaptive Priorities for Multi-Objective Alignment in Vision-Language Generation

D Chen, X Zhuang, J Xu, L Xie, Z Wang, J Zhuang, H Yang, L Dou, X He, et al.

Findings of ACL 2026 · 4922–4939

Published

Caring Without Feeling: Affective Dynamics as the Control Layer of Human-AI Agent Collaboration

Junjie Xu, Xingjiao Wu, Zihao Zhang, Yujia Xu, Yuzhe Yang, Jin Zhu, Luwei Xiao, Wen Wu, Liang He

arXiv:2606.18259 · under journal review

Preprint

Multimodal Few-Channel EEG Emotion Recognition Based on Brain Asymmetry

Q Shen, Z Zhang, J Xu, Z Lei, X Wu, D Dong, L He

ICASSP 2026

Published

Human-Centered Pathways to Trustworthy AI in Healthcare: A Comparative Analysis of Explainable AI, Human-in-the-Loop, Hybrid AI, and Uncertainty Quantification Techniques

A Kohan, **Junjie Xu**, L Xiao, X Wu, A Kukkar, S Hussain, M Roshanzamir, R Alizadehsani, UR Acharya

Research Square preprint · under journal review

Preprint

MARS: Multimodal-Assisted Refined Semantic Alignment

Junjie Xu, Xingjiao Wu, Zihao Zhang, Shuwen Yang, Tianlong Ma, Daoguo Dong, Liang He

Information Processing & Management 63 (1), 104292

Published

Large Language Models have Chain-of-Affective

Junjie Xu, Xingjiao Wu, Luwei Xiao, Yuzhe Yang, Jie Zhou, Zihao Zhang, Luhan Wang, Yi Huang, Nan Wu, Yingbin Zheng, Chao Yan, Cheng Jin, Honglin Li, Liang He

arXiv:2512.12283

Preprint

Temporal-Conditioned Symbolic Alignment for Controllable Text-to-Music Generation

Z Zhang, X Wu, J Xu, et al.

ACM Multimedia 2025 · 10728–10737

Published

Multi-Type Preference Learning: Enabling Minimum-Redundancy Maximum-Relevance Preference Combination for Surgical Gesture Recognition

Z Liu, J Xu, et al.

ICRA 2025 · 1163–1169

Published

Bidirectional Directed Acyclic Graph Neural Network for Aspect-level Sentiment Classification

Junjie Xu, Luwei Xiao, Anran Wu, Tianlong Ma, Daoguo Dong, Liang He

ACM Transactions on Asian and Low-Resource Language Information Processing

Published

Aesthetic Matters in Music Perception for Image Stylization: An Emotion-Driven Music-to-Visual Manipulation

Junjie Xu, Xingjiao Wu, T Yao, Zihao Zhang, J Bei, W Wen, Liang He

arXiv:2501.01700

Preprint

Attention Mixture Network for Crowd Counting via Binarization Transfer

Junjie Xu, Zihao Zhang, Xin Li, Weijie Li, Kun Yu

McGE Workshop · ACM Multimedia 2024 Published

Atlantis: Aesthetic-Oriented Multiple Granularities Fusion Network for Sentiment Analysis

Luwei Xiao, Xingjiao Wu, Junjie Xu, Weijie Li, Cheng Jin, Liang He

Information Fusion 106, 102304 Published

EmoStyle: Emotion-Aware Semantic Image Manipulation with Audio Guidance

Q Shen, Junjie Xu, J Mei, X Wu, D Dong

Applied Sciences 14 (8), 3193 Published

Simple Contrastive Learning in a Self-Supervised Manner for Robust Visual Question Answering

S Yang, L Xiao, X Wu, Junjie Xu, L Wang, L He

Computer Vision and Image Understanding 241, 103976 Published

UMAAF: Unveiling Aesthetics via Multifarious Attributes of Images

W Li, Y Wan, X Wu, Junjie Xu, C Jin, Liang He

IEEE TCSVT (under review) · arXiv:2311.11306 Manuscript

Cross-Modal Fine-Grained Alignment and Fusion Network for Multimodal Aspect-Based Sentiment Analysis

Luwei Xiao, Xingjiao Wu, Shuwen Yang, Junjie Xu, J Zhou, Liang He

Information Processing & Management 60 (6), 103508 Published

Automatic Image Aesthetic Assessment for Human-Designed Digital Images

Y Wan, W Li, X Wu, Junjie Xu, J Yang

McGE Workshop · ACM Multimedia 2023 Published

DCQA: Document-Level Chart Question Answering Based on Multimodal Semantic Understanding

A Wu, L Xiao, X Wu, S Yang, Junjie Xu, Z Zhuang, N Xie, C Jin, L He

SSRN 4703693 Preprint

BPGG: Bidirectional Prototype Generation and Guidance Network for Few-Shot Anomaly Localization

J Zhang, Z Zhuang, Junjie Xu, T Ma, Liang He

ICANN 2022 · 24–36 Published

Graph Convolution over the Semantic-Syntactic Hybrid Graph Enhanced by Affective Knowledge for Aspect-Level Sentiment Classification

Junjie Xu, Shuwen Yang, Luwei Xiao, Zhichao Fu, Xingjiao Wu, Tianlong Ma, Liang He

IJCNN 2022 Published

Phoenix: Prioritize the Hierarchical Observation of Auditory Expression for Audio-Based Image Generation

Junjie Xu, Xingjiao Wu, Zihao Zhang, Luwei Xiao, Daoguo Dong, Liang He

IEEE TCSVT (under review) · SSRN 5343200 Manuscript

COFFEE: Listen, with Your Eyes: Cross-Modal Fine-Grained Alignment and Fusion Enhanced for Audio-Based Image Generation

Junjie Xu and collaborators

IEEE TNNLS (under review)

Manuscript

Cross-Modal Alignment and Human Preference Learning for Fine-Grained Music-Guided Image Generation

Junjie Xu and collaborators

Under review · ICLR 2026

Under review

ElysianMV: Bridging Therapeutic Emotions and Aesthetic Imagery via Music-Visual Cross-Modal Synthesis

Junjie Xu, Xingjiao Wu, Zihao Zhang, Luwei Xiao, Jiabao Zhao, Liang He

Preprint submitted to Pattern Recognition

Preprint

GARBO-T2I: Gradient-Aware Reinforcement-Based Optimization for Text-to-Image Generation

Junjie Xu and collaborators

Research manuscript

Manuscript

HealBench: A Fine-Grained Benchmark for Evaluating Psychological Comfort in Text-to-Image Generation

Junjie Xu and collaborators

Research manuscript

Manuscript

Evidence Before Severity: Auditing Safety-Evidence Fidelity and Metadata Sensitivity in Depression-Related Dialogues

Junjie Xu and collaborators

Under review · AAAI 2027

Under review

Affective Dynamics in Human–AI Agent Collaboration: A Survey of Mechanisms, Risks, and Governance

Junjie Xu, Xingjiao Wu, Zihao Zhang, Yujia Xu, Yuzhe Yang, Jin Zhu, Luwei Xiao, Wen Wu, Liang He

Under review · ACM Computing Surveys

Under review

PATENTS

All patents filed as co-inventor; publication numbers verifiable on China's patent gazette.

一种基于Kapur阈值法的黑色素瘤皮肤镜图像分割方法
CN 118840378 A · 2024-10-25

一种基于人在回路与视觉基础模型的医学图像增强方法
CN 120298246 A · 2025-07-11

一种基于人在回路中标注样本的医学图像分割方法
CN 118485681 A · 2024-08-13

一种基于偏好传递性的人在回路强化学习数据增强方法
CN 118916133 A · 2024-11-08

一种基于图像多属性建模的美学评估方法
CN 117422962 A · 2024-01-19

一种基于多层次对抗学习的图像质量评估方法
CN 120198411 A · 2025-06-24

一种基于多模态交叉注意力与动态特征融合的音乐生成方法
CN 120316301 A · 2025-07-15

一种基于对抗神经网络的音频驱动图像编辑方法
CN 118524266 A · 2024-08-20

一种基于情感驱动和人类专家反馈的机器行为学习方法
CN 120318634 A · 2025-07-15

一种基于情感驱动的多乐器分离协同音乐生成方法
CN 120319204 A · 2025-07-15

一种基于扩散模型的小样本字体生成方法
CN 118898549 A · 2024-11-05

一种基于有向无环图网络的方面级情感分类方法
CN 116127072 A · 2023-05-16

一种基于注意力图卷积网络的多模态融合机械缺陷检测方法
CN 114170477 A · 2022-03-11

基于人在回路与视觉基础模型的少样本医学图像分割方法
CN 120298698 A · 2025-07-11

RESEARCH PROJECTS

Multimodal Digital Therapeutics [Project Director](#)

Interdisciplinary Research Project · ECNU · 2024 — 2025 · Rated Excellent

Led a research project on multimodal digital support, emotional experience, and human-centered interaction.

Emotion-Controllable Music-to-Image [Project Director](#)

Doctoral Research Innovation Fund · ECNU · 2023 · Advised by Liang He

Explored how human feedback and affective intent shape cross-modal music-to-image generation.

EduChat-R1 [Participant](#)

Open Education Reasoning Model · ECNU · 2025 · Open-source release

Contributed to an open education reasoning model through research, engineering, and open-source collaboration.

HONORS & AWARDS

2025	Kuanrui Talent Scholarship ECNU School of Computer Science and Technology
2023	Shanghai Outstanding Graduate Shanghai Municipal Education Commission
2023	National Third Prize · Challenge Cup China Association for Science and Technology
2022	Huaxin Scholarship ECNU School of Computer Science and Technology